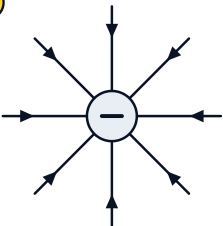
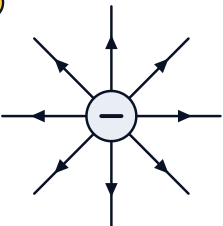


Static charge and electric fields

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 8 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	Describe how insulating materials can become electrically charged.	By rubbing them together: friction transfers electrons between them.	☐
2 ★	Describe how a material can become negatively charged.	By gaining electrons.	☐
3	Describe how a material can become positively charged.	By losing electrons.	☐
4 ★	Describe what happens when two charged objects are brought close together.	They exert a force on each other: like charges repel, opposite charges attract.	☐
5	State whether electrostatic force is a contact force or non-contact force.	Non-contact force	☐
6	Explain why there is no potential difference between two insulators before they are rubbed together.	Both are neutral, so there is no difference in charge.	☐
7	Name the type of field that surrounds every charged object.	An electric field	☐
8	State where the electric field around a charged object is strongest.	Closest to the object.	☐
9	Describe how the strength of an electric field changes with distance from the charge.	The field gets stronger closer to the charge.	☐
10 ★	State the direction in which electric field lines point.	Away from positive charges and towards negative charges.	☐

	QUESTION	ANSWER	✓
11	Which diagram, A or B, shows the electric field around a negative charge? Explain. A  B 	A. The field lines point towards the negative charge.	☐
12	Explain why a charged object experiences a force when close to another charged object.	The electric field of each object exerts a force on the other object.	☐
13	State how the size of the force between two charged objects can be increased.	By moving them closer together.	☐
14	State what happens when two objects with the same type of charge are brought together.	They repel each other.	☐
15	State what happens when two objects with opposite charges are brought together.	They attract each other.	☐
16 ★	A plastic rod becomes negatively charged when it is rubbed with a cloth. State and explain the charge on the cloth.	Positive (equal in size): the cloth lost the electrons that the rod gained.	☐
17 ★	What is an electric field?	The region around a charged object where another charged object experiences a force.	☐
18 ★	Describe the electric field around an isolated positively charged sphere.	Straight lines pointing radially outwards, away from the sphere.	☐
19 ★	Explain how a spark is produced between a charged object and an earthed conductor.	The strong electric field ionises the air, so charge can flow across the gap.	☐